Evaluating Multilingual Relevance in LongEval

Team SearChii

Web Information Retrieval SoSe 2026
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The **problem** we studied

LANGUAGE BARRIER IN RELEVANCE JUDGING

LANGUAGE BARRIER

English-speaking judges cannot assess non-English papers

HIDDEN RELEVANCE

Some excluded papers are actually highly relevant

AUTOMATIC EXCLUSION

Unreadable documents are often labeled as irrelevant (0)

BIASED EVALUATION

Retrieval metrics become distorted for multilingual systems

Research Question & **Hypotheses**

RESEARCH QUESTION

Does translating non-English documents that were originally judged as irrelevant into English change their relevance judgements, and do these re-judgements differ between human judges and an LLM-based judge?

HYPOTHESIS 1

Translating previously irrelevant non-English documents into English increases the proportion of documents receiving a relevant judgement (≥1)

HYPOTHESIS 2

Human and LLM-based relevance judgements significantly differ for translated non-English documents

Data Filtering **Process**

1

Filter Qrels

- Selected initial rel = 0 docs
- 688 raw documents identified

2

Match corpus

- Linked IDs with LongEval-Sci corpus
- Retrieved titles and abstracts

3

Detect languages

- Applied langdetect to abstracts
- Filtered non-English docs

4

Split dataset

- Split docs into 3 batches
- Prepared assessment

5

Translate

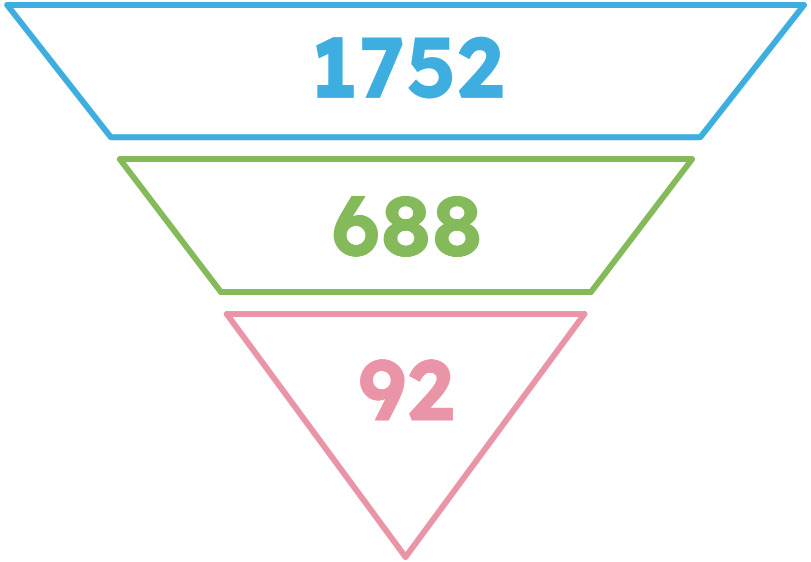
- Translated docs with DeepL
- Extracted structural metadata

6

Relabel

- Team: Manual reassessment
- LLM: Zero-shot via Mistral Medium

Corpus **Overview**



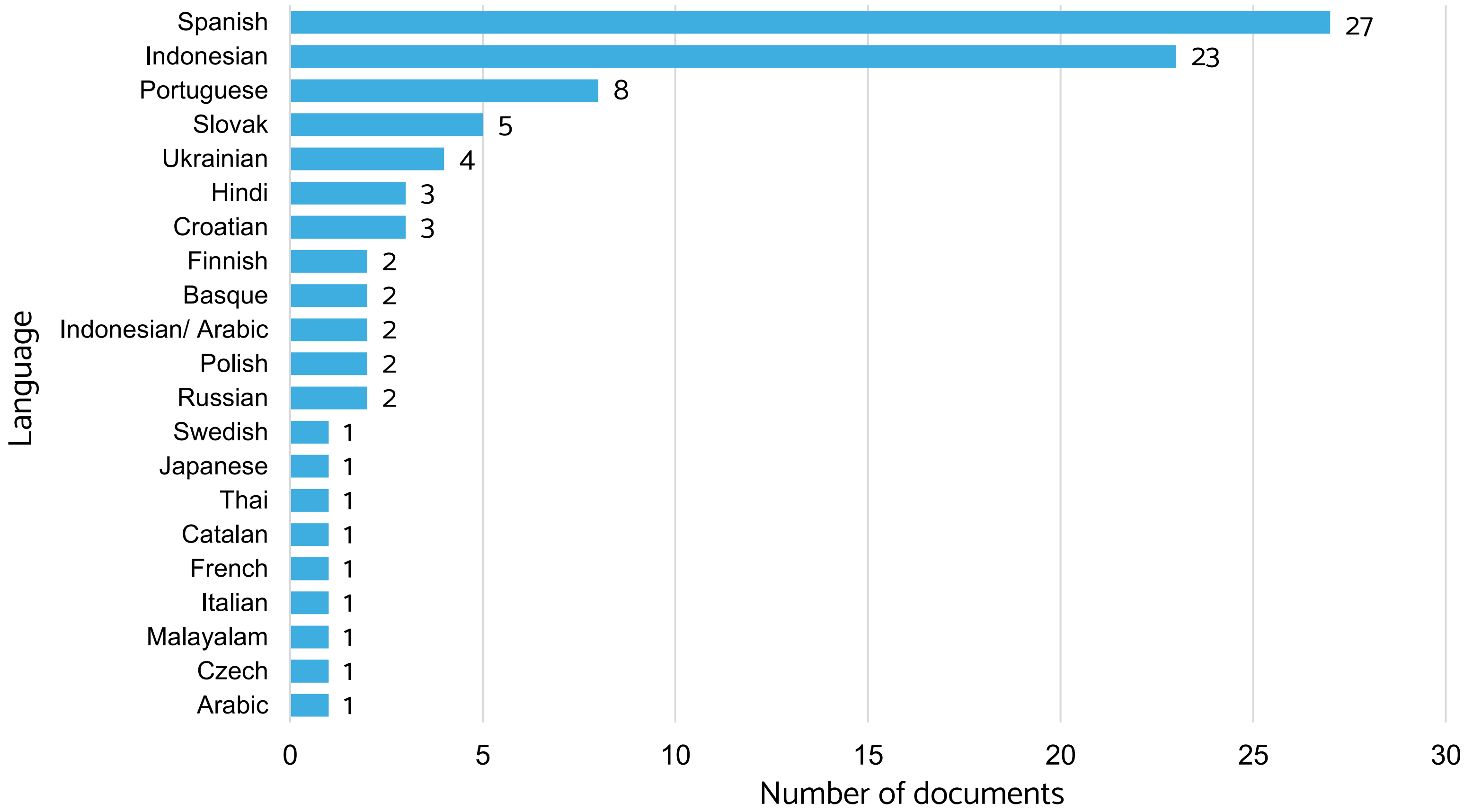
Starting point: **1752 judged documents**

688 documents were identified with an initial relevance of 0 (irrelevant) and available full text

Language filtering identified **92 non-English documents**

These documents formed the basis for translation and relabelling

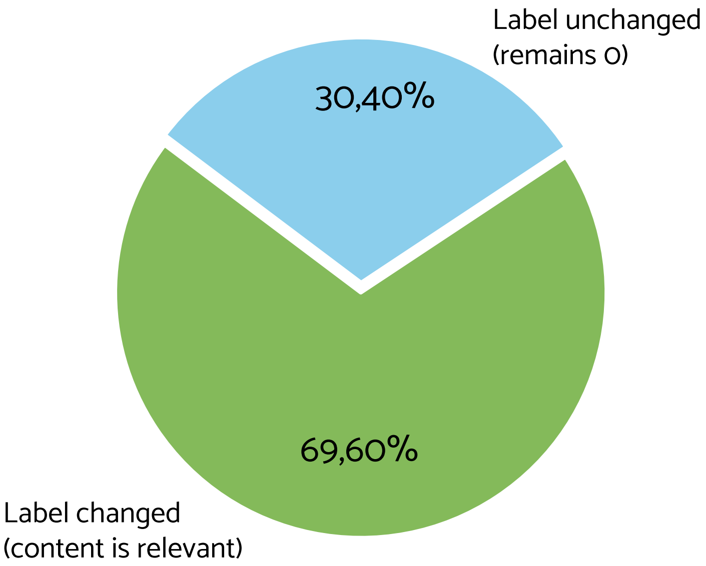
Language Distribution of the Relabelled Documents



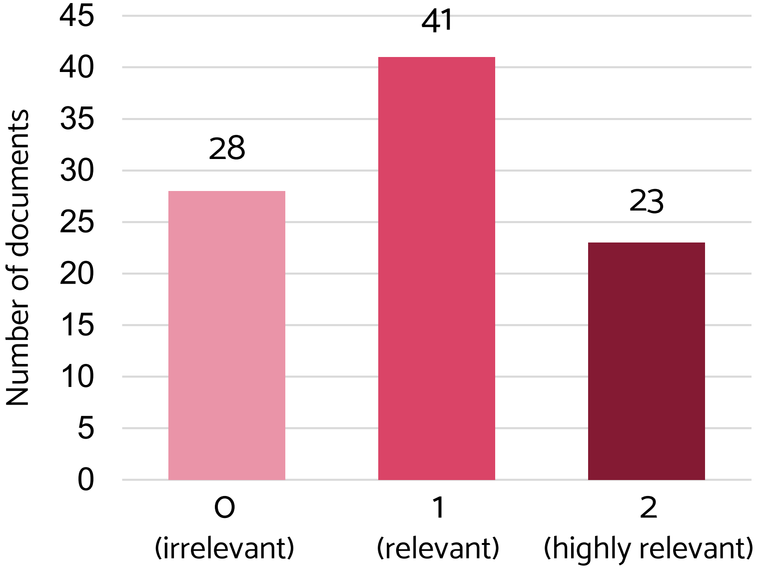
Impact of **Human** Relabelling

Starting point: 92 non-English documents originally labelled as irrelevant (rel = 0)

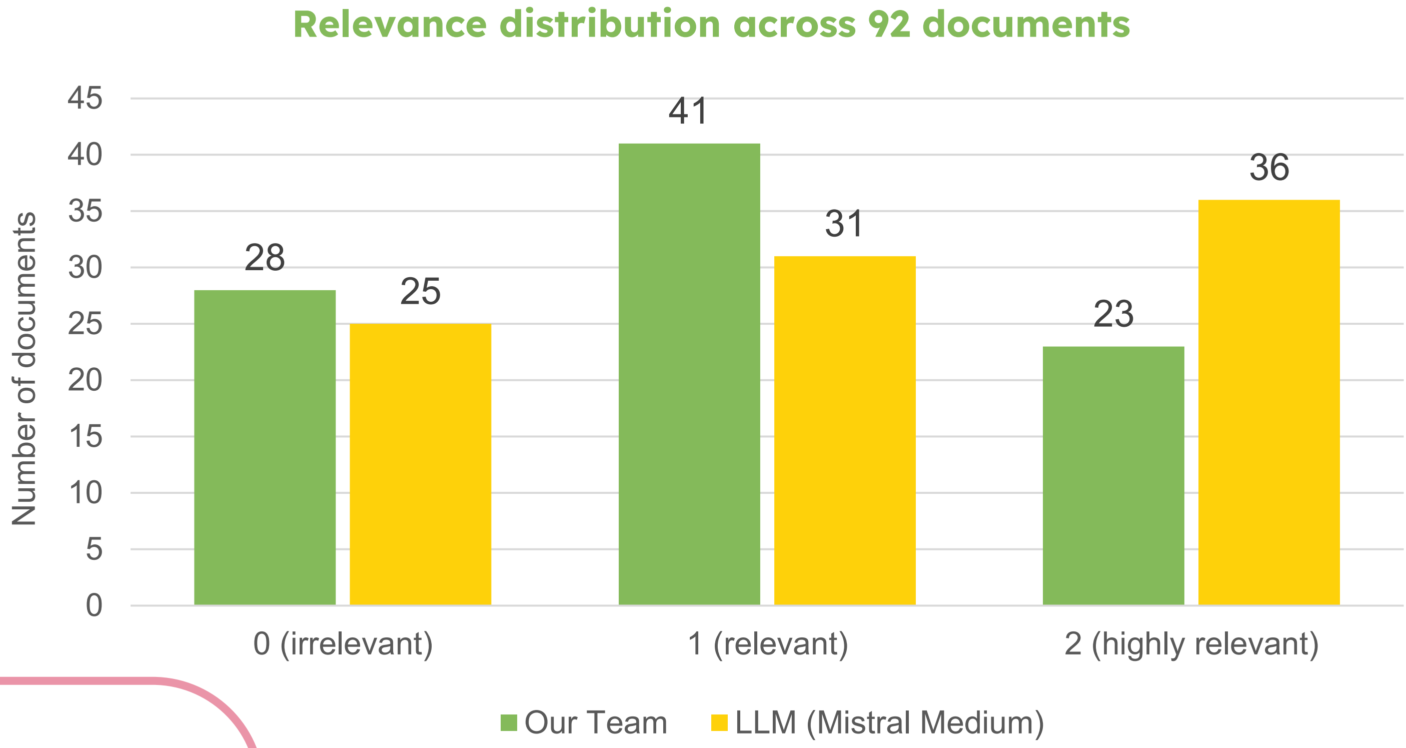
Did Judgements Change?



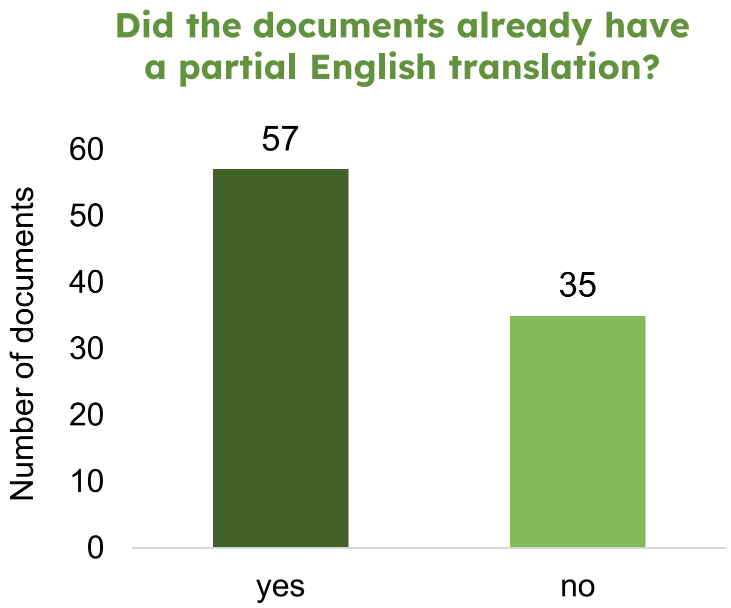
Final Relevance Distribution

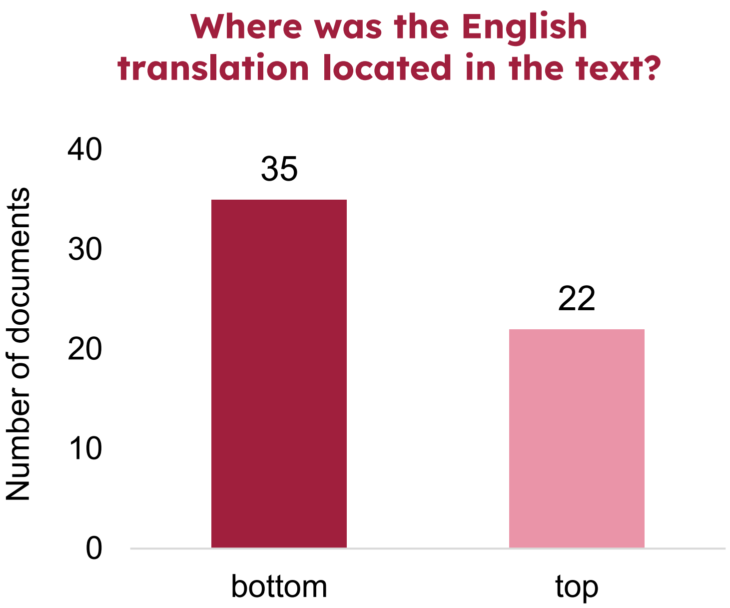


Overall Human–LLM **Comparison**



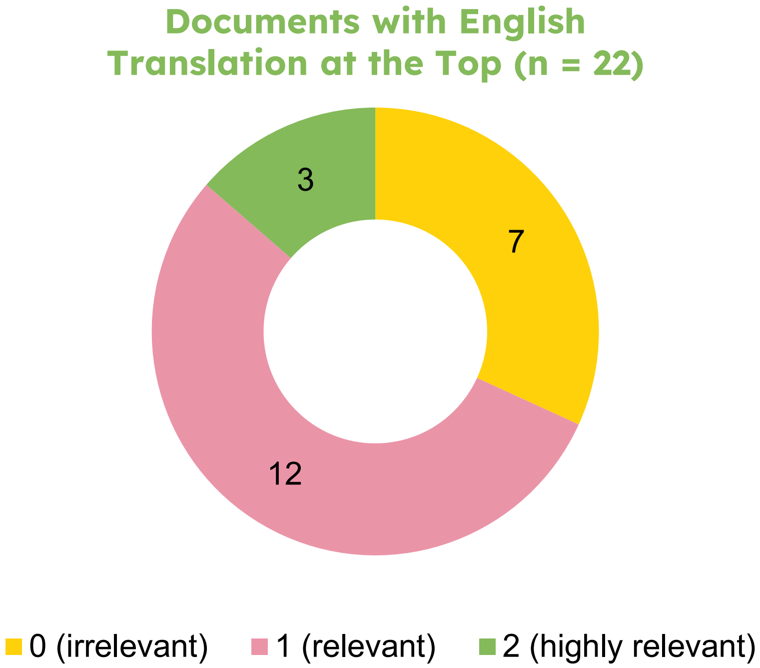
English Translation **Availability**





62% of foreign documents contained english text, but **61%** placed it at the bottom, which may cause judges to skip them

Does Translation **Position** Matter?



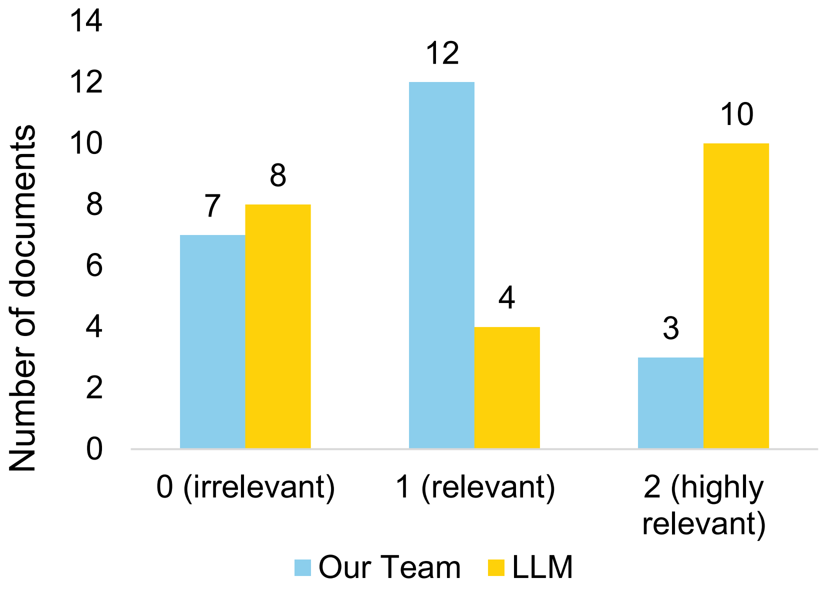
- **22** documents had English abstracts available at the top, yet all were judged irrelevant
- **15/22** documents (68%) were reclassified as relevant by us
- **16/22** documents (73%) still had non-English titles
- Foreign-language titles may have influenced first impressions before the english abstract was considered

Judgement **Comparison** (n=22)

- Humans mostly assigned ”Relevant”
- LLM assigned three times more ”Highly Relevant” labels
- Both agreed on the number of irrelevant documents
- Main disagreement lies between 1 vs. 2

LLM classified 14/22 documents (64%) as relevant or highly relevant

How did the LLM rate these 22 documents?



Why did the LLM assign so many "2"s?

EXAMPLE: BIOBATTERY USING BANANA PEELS

NARRATIVE

How to use banana peels to build a biobattery and generate electrical power

DOCUMENT

Biogas production from pig manure + poultry manure + banana peels

RATING

Human → 1 (relevant)
LLM → 2 (highly relevant)

ANALYSIS INSIGHTS

- document matches keywords but differs in the actual method
- LLM rated paper based on this strong topical overlap
- Humans → 1, as the study focuses on biogas, not biobatteries
- main disagreement is relevance depth (1 vs. 2)

Key **Findings** from the Analysis

Human Relabelling

69.6% of documents (64/92) received a new relevance label

Language Bias

Humans overlooked relevant text when titles were non-English

Human vs. LLM

Main disagreement was between Relevant (1) and Highly Relevant (2)

Relevance Depth

Humans judged relevance more conservatively; LLM assigned more „Highly Relevant“ labels

Human Validation

Humans better distinguished conceptual relevance from simple keyword overlap

Evaluation Setup & Metrics

EVALUATION PIPELINE

Framework: PyTerrier

System evaluated:
BM25 vs. Dense Retrieval

3 Qrels sets:
Baseline (original)
Human (re-judged)
LLM (Mistral Medium)

METRICS INCLUDED

Precision@k, **Recall@k**, **F1@k**

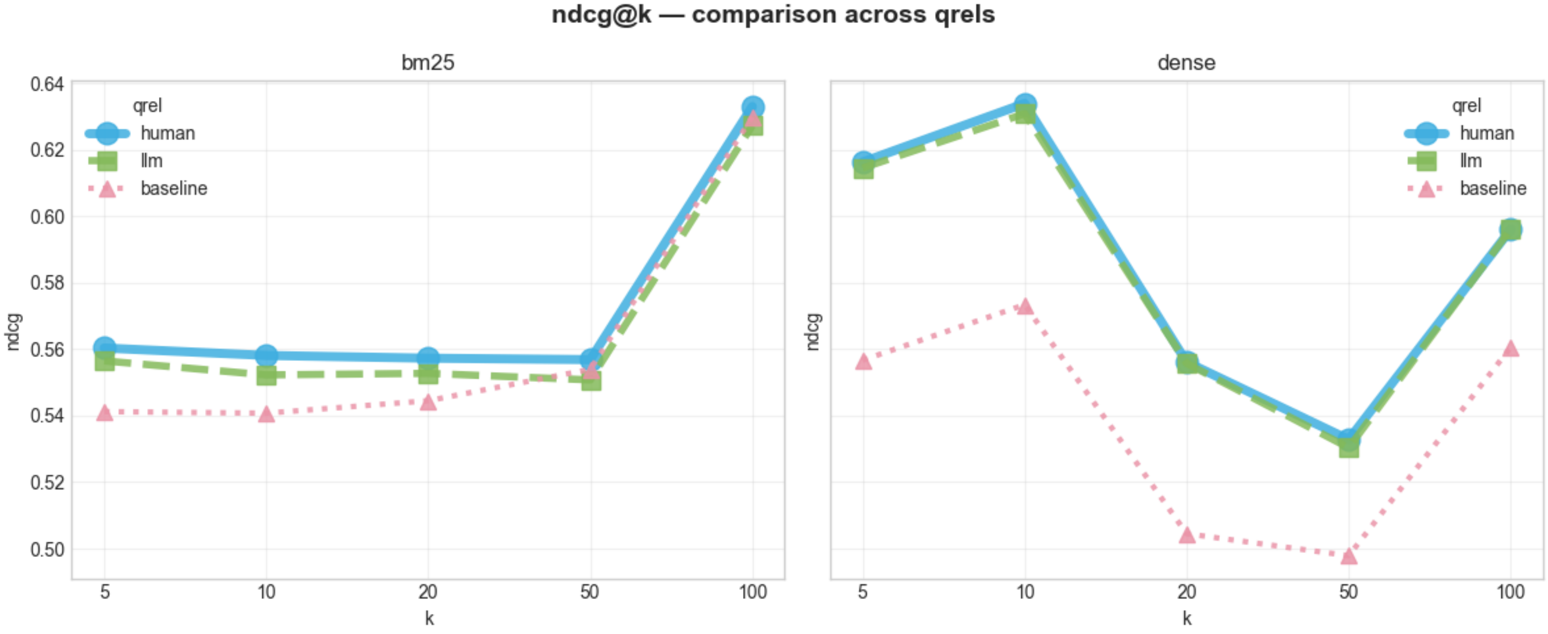
MAP@k
(Mean Average Precision)

nDCG@k
(Normalized Discounted Cumulative Gain)

MRR
(Mean Reciprocal Rank)

Overall Retrieval Impact (**nDCG@k**)

nDCG@k – comparison across qrels



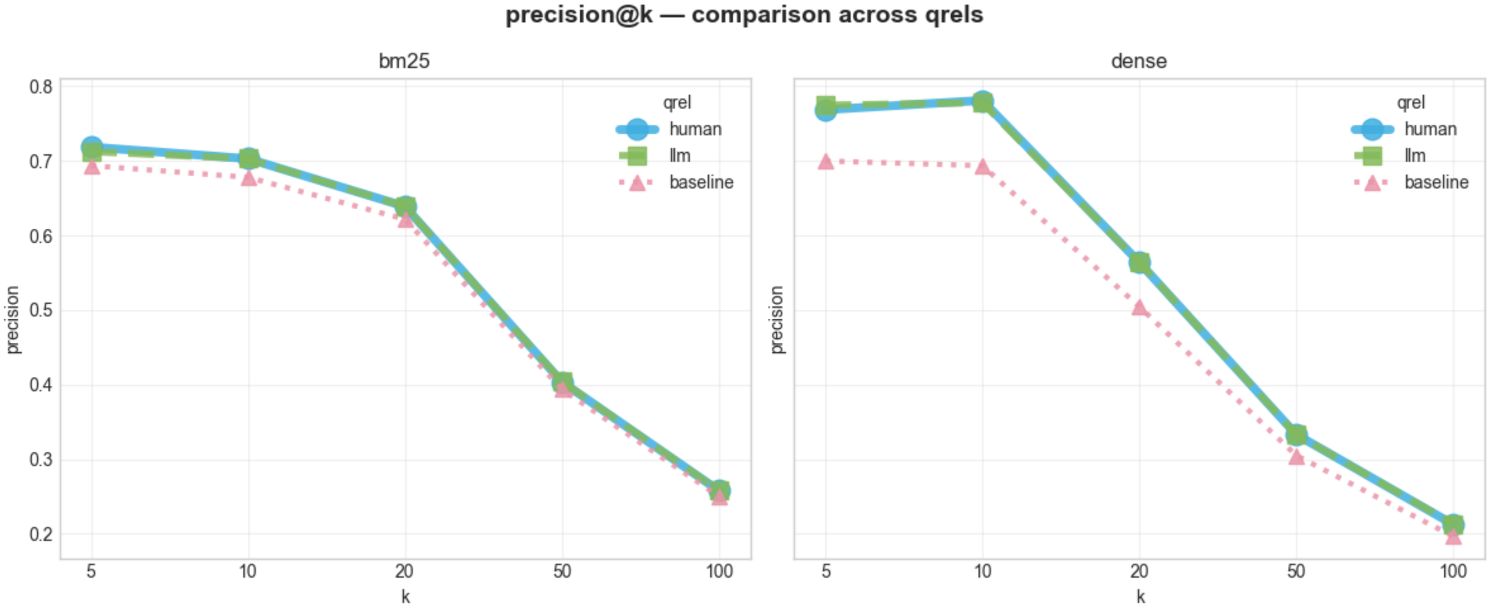
Human = LLM: human & llm curves are nearly identical

BM25 less Change: Less likely to return non-english documents in higher rank

Dense Constant Change: More likely to retrieve other languages.

Precision Analysis

precision@k – comparison across qrels



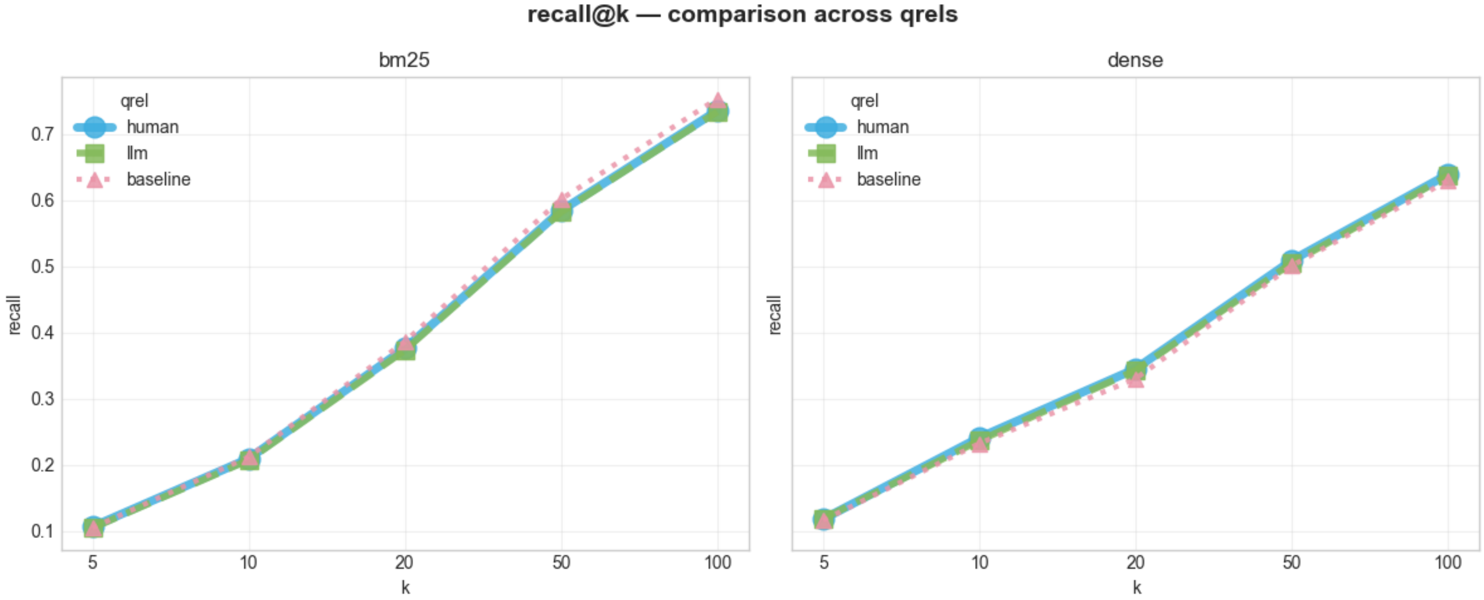
Dense Retrieval: Precision@5 rises from 0.70 → 0.78 (human/llm)

BM25: Minimal changes across all k

Key Insight: Dense model retrieves relevant non-English docs more

Recall Analysis

recall@k – comparison across qrels



Recall Unchanged: All three lines overlay completely across k=5 to k=100

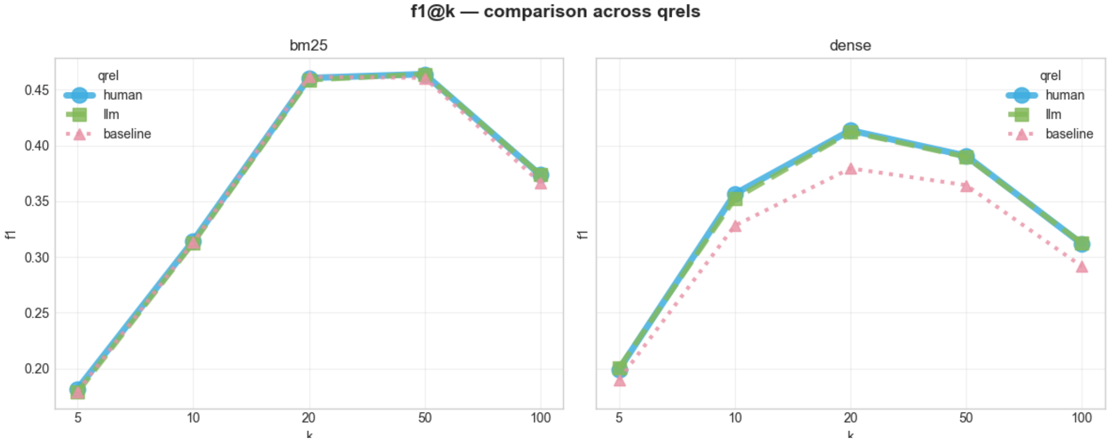
Meaning: The models had already captured these non-English documents

Key Insight: Models already found the docs; translation just fixed "false negative" penalty

MAP@k and F1@k

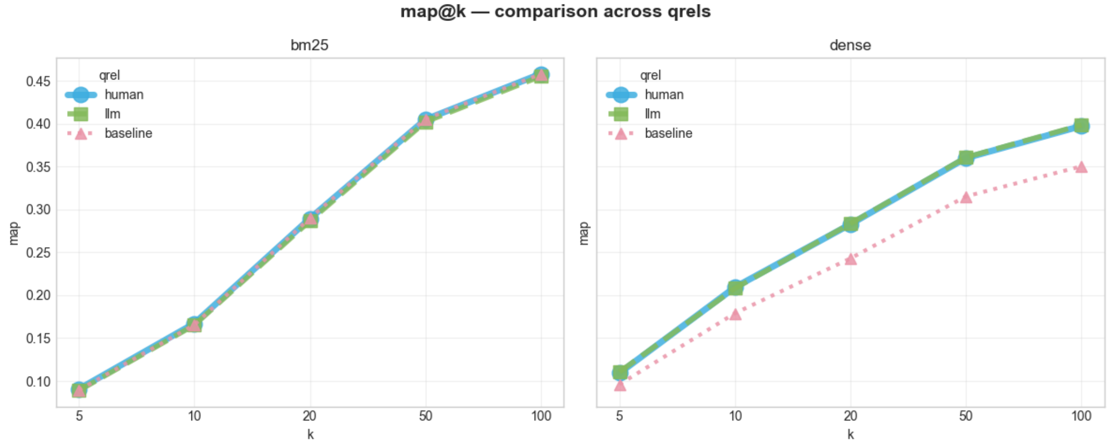
f1@k – comparison across qrels

- Peaks at k=20
- Dense retrieval gains the most between k=10 and k=50



map@k – comparison across qrels

- Clear lift for Dense Retrieval (MAP@100 increases from 0.35 to 0.40)
- BM25 performance remains completely unchanged



Evaluation Takeaways

Similar Human & LLM Results

Different label ratings had very little impact on the overall system ranking

Dense Retrieval Benefited Most

Dense retrieval gained the largest improvements from the updated qrels

Small Overall Effect

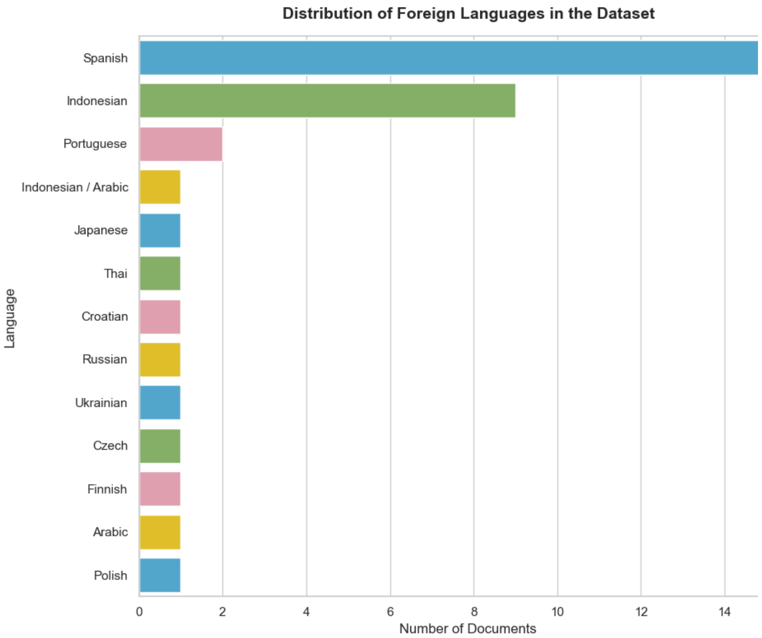
Relabelling 92 documents reduced test collection bias without changing overall rankings

Thank you for your attention!

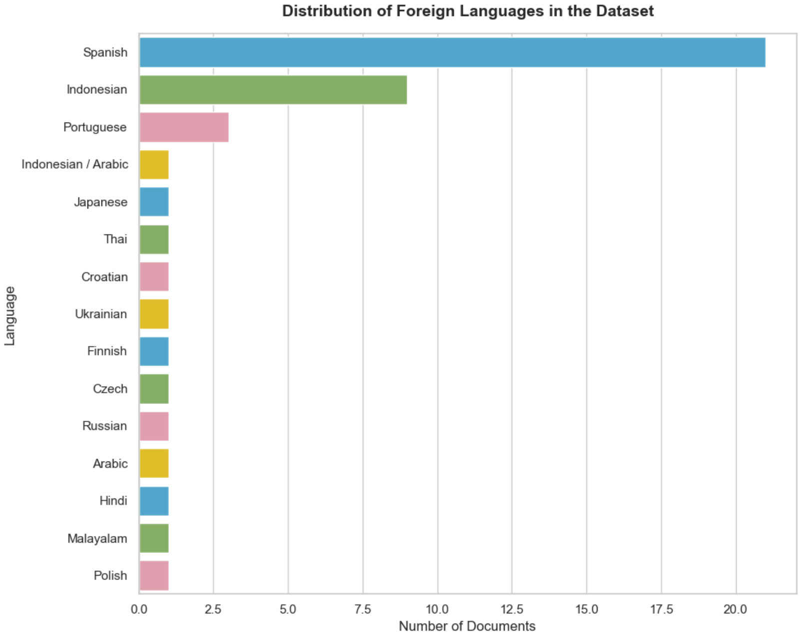
Any questions?

Some Extras

BM25

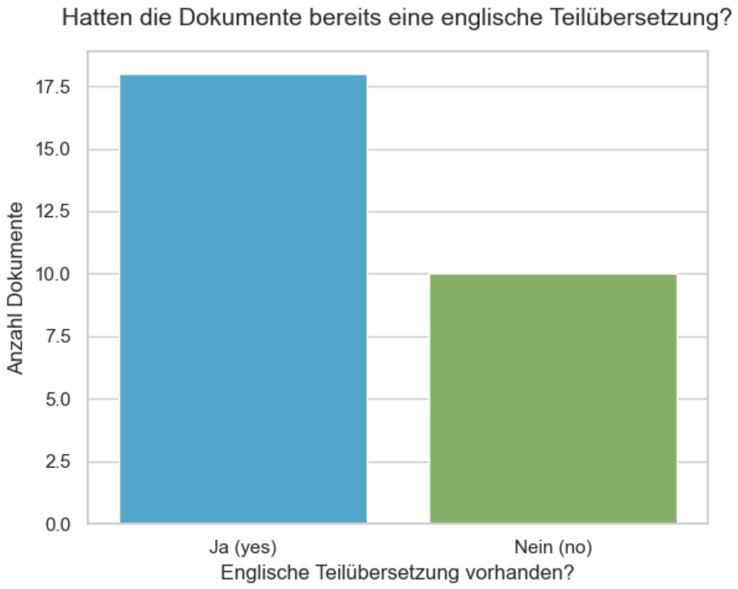


Dense

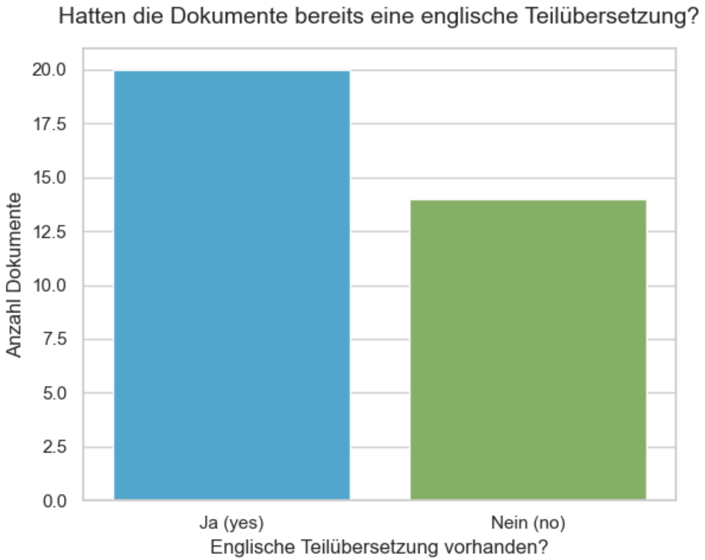


Some Extras

BM25

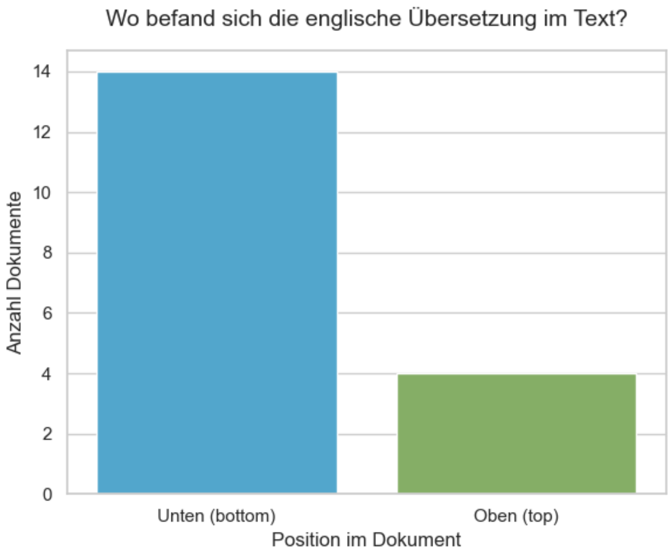


Dense



Some Extras

BM25



Dense